WEEK 6: PRODUCT REVIEW AND RATING MANAGEMENT SYSTEM

SCENARIO:

The e-commerce company wants to collect customer feedback about purchased products. Customers can provide ratings and reviews after completing their orders. The system should store customer opinions, analyze product performance, and identify highly rated products.

Students will create a Product Review and Rating Module connected with the existing Customer and Product modules.

1. Create Review and Rating Table

```
SQL> CREATE TABLE Review (
  2      ReviewID VARCHAR2(7),
  3      CustomerID NUMBER,
  4      ProductID VARCHAR2(7),
  5      Rating NUMBER(1),
  6      ReviewText VARCHAR2(50),
  7      ReviewDate DATE,
  8      PRIMARY KEY (ReviewID),
  9      FOREIGN KEY (CustomerID) REFERENCES Customer(Customer_ID),
 10      FOREIGN KEY (ProductID) REFERENCES Product(Product_ID)
 11  );

Table created.

SQL> desc review
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 REVIEWID                                  NOT NULL VARCHAR2(7)
 CUSTOMERID                                         NUMBER
 PRODUCTID                                          VARCHAR2(7)
 RATING                                             NUMBER(1)
 REVIEWTEXT                                         VARCHAR2(50)
 REVIEWDATE                                         DATE
```

Store customer feedback and ratings

Tasks:

1. Add reviews for purchased products.

```
SQL> INSERT INTO Review VALUES ('REV201', 2, 'P001', 5, 'Excellent quality', DATE '2026-10-10');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES ('REV202', 3, 'P002', 4, 'Good product', DATE '2026-10-11');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES ('REV203', 4, 'P003', 3, 'Average performance', DATE '2026-10-12');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES ('REV204', 5, 'P004', 2, 'Battery needs improvement', DATE '2026-10-13');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES ('REV205', 6, 'P005', 1, 'Not satisfied', DATE '2026-10-14');

1 row created.
```

```
SQL> SELECT * FROM Review;

REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEXT                                         REVIEWDAT
-------------------------------------------------- ---------
REV201           2 P001              5
Excellent quality                                  10-OCT-26

REV202           3 P002              4
Good product                                       11-OCT-26

REV203           4 P003              3
Average performance                                12-OCT-26


REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEXT                                         REVIEWDAT
-------------------------------------------------- ---------
REV204           5 P004              2
Battery needs improvement                          13-OCT-26

REV205           6 P005              1
Not satisfied                                      14-OCT-26
```

2. Store customer ratings from 1 to 5 stars.

Ensure this by adding Constraint

```
SQL> ALTER TABLE Review
  2  ADD CONSTRAINT CHK_REVIEW_RATE
  3  CHECK (Rating >= 1 AND Rating <= 5);

Table altered.
```

```
SQL> INSERT INTO Review
  2  VALUES ('REV206', 3, 'P002', 5,
  3          'Excellent performance', DATE '2026-10-20');

1 row created.
```

3. Update review comments if required

```
SQL> UPDATE Review
  2  SET ReviewText = 'Good quality but needs improvement'
  3  WHERE ReviewID = 'REV203';

1 row updated.
```

4. Remove inappropriate or invalid reviews.

```
SQL> DELETE FROM Review
  2  WHERE ReviewID = 'REV206';

1 row deleted.
```

Retrieve Product Review Details

Write a SQL queries to:

1. Display all reviews for a product.

```
SQL> SELECT * FROM Review
  2  WHERE ProductID = 'P005';

REVIEWI CUSTOMERID PRODUCT     RATING
------- ---------- ------- ----------
REVIEWTEXT                                                 REVIEWDAT
---------------------------------------------------------- ---------
REV205           6 P005              1
Not satisfied                                              14-OCT-26


SQL>
```

2. Display customer name with their reviews.

```
SQL> SELECT C.CUSTOMER_NAME, R.PRODUCTID,
  2         R.RATING, R.REVIEWTEXT, R.REVIEWDATE
  3  FROM Customer C
  4  INNER JOIN Review R
  5  ON C.CUSTOMER_ID = R.CUSTOMERID;

CUSTOMER_NAME   PRODUCT     RATING
--------------- ------- ----------
REVIEWTEXT                                                 REVIEWDAT
---------------------------------------------------------- ---------
Arun            P001             5
Excellent quality                                          10-OCT-26

Priya           P002             4
Good product                                               11-OCT-26

Karthik         P003             3
Good quality but needs improvement                         12-OCT-26


CUSTOMER_NAME   PRODUCT     RATING
--------------- ------- ----------
REVIEWTEXT                                                 REVIEWDAT
---------------------------------------------------------- ---------
Divya           P004             2
Battery needs improvement                                  13-OCT-26

Rahul           P005             1
Not satisfied                                              14-OCT-26
```